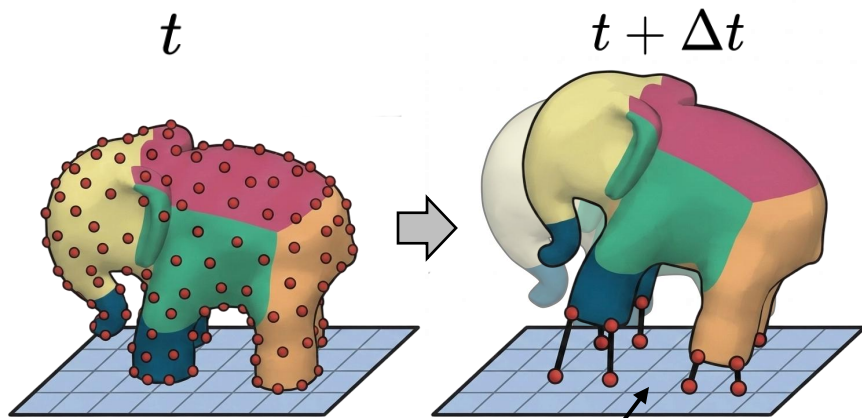
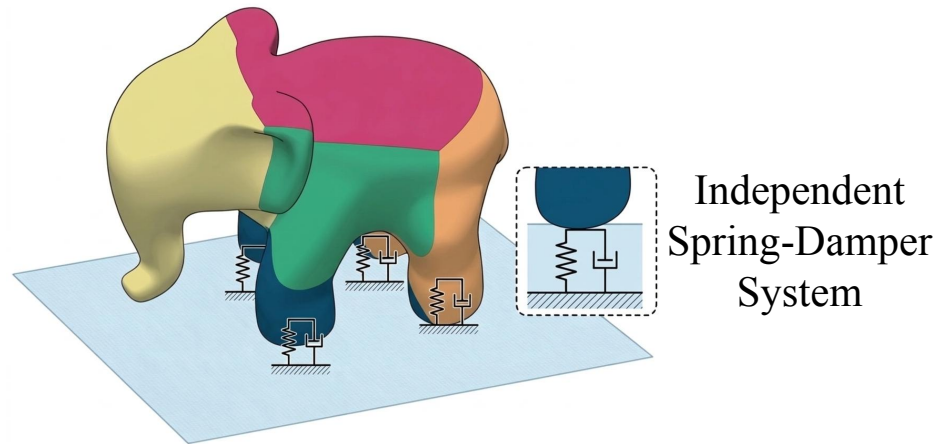


Assumption 1: Fixed Contact Set



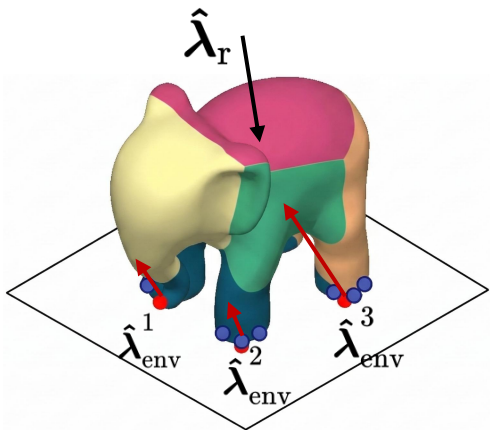
Contacts remain unchanged over a short time interval (Sticking).

Assumption 2: Multi-Contact Decoupling



Neglecting the coupling among environment-object contacts

Surrogate Contact Model



$$\hat{\mathbf{p}}_r \in \mathcal{I}_{\text{avail}}, \hat{\boldsymbol{\lambda}}_r \in \mathcal{K}$$

$$\hat{\boldsymbol{\lambda}}_{\text{env},i} = \max \left(-\mathbf{D}^{-1} \tilde{\mathbf{J}}_{\text{env},i} \mathbf{M}_o^{-1} (\boldsymbol{\tau}_o + \sum_{i=1}^{n_r} \mathbf{J}_{r,i}^\top \hat{\boldsymbol{\lambda}}_{r,i}), \mathbf{0} \right).$$

$$f_{\text{SCM}} : \begin{cases} \hat{\mathbf{v}}^+ = \frac{1}{h} \mathbf{M}_o^{-1} \tilde{\mathbf{b}}_o + \frac{1}{h} \mathbf{M}_o^{-1} \sum_{i=1}^{n_{\text{env}}} \tilde{\mathbf{J}}_{\text{env},i}^\top \hat{\boldsymbol{\lambda}}_{\text{env},i}, \\ \hat{\mathbf{x}}_{k+1} = \mathbf{x}_k + h(\mathbf{v}_k + \hat{\mathbf{v}}^+). \end{cases}$$

SCM is a lightweight approximation of the object-environment contacts. With SCM, the environment contact force is an explicit function of robot contact force.